

Quick Start & POC Guide

From zero to migrated VM in 30 minutes. Give this to your IT team.

Part 1: Install (5 minutes)

1 Prerequisites

Any Linux machine (Fedora, RHEL 9+, Ubuntu 22.04+). 4GB RAM, 20GB free disk. Root access.

```
sudo dnf install -y qemu-img libguestfs-tools # Fedora/RHEL
# OR
sudo apt install -y qemu-utils libguestfs-tools # Ubuntu/Debian
```

2 Install hyper2kvm

```
git clone https://github.com/ssahani/hyper2kvm.git
cd hyper2kvm
pip3 install -e .
# OR use the quickstart script:
sudo ./scripts/quickstart.sh
```

3 Install govc (for vSphere)

```
curl -L https://github.com/vmware/govmomi/releases/latest/download/govc_Linux_x86_64.tar.gz \
| sudo tar xzf - -C /usr/local/bin govc
```

4 Verify

```
h2kvmctl --version # should print version
govc version       # should print govc version
```

Part 2: Local Test (10 minutes)

Test with a local VMDK file — no vCenter needed.

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Get a test VMDK 2 min

Use any existing VMDK, or download a small test image:

```
# Option A: Use an existing VMDK from your VMware datastore
# Option B: Download Photon OS (small, fast)
curl -LO https://packages.vmware.com/photon/5.0/Rev1/ova/photon-5.0-ova.ova
tar xf photon-5.0-ova.ova # extracts .vmdk
```

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Convert VMDK to qcow2 5 min

```
sudo h2kvmctl \
  --cmd local \
  --vmdk photon.vmdk \
  --output-dir ./out \
  --out-format qcow2 \
  --flatten \
  --fstab-mode stabilize-all \
  --regen-initramfs \
  --emit-domain-xml \
  --vm-name photon-test \
  --memory 1024 \
  --vcpus 2
```

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Verify output 1 min

```
ls -lh out/           # qcow2 file created
qemu-img info out/*.qcow2 # verify format
virsh list --all       # VM should be defined
```

Success criteria: qcow2 file created, VM defined in libvirt, `virsh start photon-test` boots successfully.

Part 3: vSphere Test (15 minutes)

Connect to your vCenter and migrate a real VM.

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Set vCenter credentials 1 min

```
export GOVC_URL="https://vcenter.yourdomain.com/sdk"
export GOVC_USERNAME="administrator@vsphere.local"
export GOVC_PASSWORD="your-password"
export GOVC_DATACENTER="your-datacenter"
export GOVC_INSECURE=1 # if self-signed cert
```

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Discover VMs 1 min

```
# Quick check:
govc ls /your-datacenter/vm/

# Or use the interactive TUI:
./zkvm/zkvm
# Press Tab to go to vSphere tab – VMs auto-discovered
```

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Migrate a test VM 10-30 min

```
# Create migration config:
cat > test-migration.yaml <<EOF
cmd: vsphere
vcenter: vcenter.yourdomain.com
vc_user: administrator@vsphere.local
vc_password_env: GOVC_PASSWORD
dc_name: your-datacenter
vs_action: export_vm
vs_vm: your-test-vm
output_dir: ./out
out_format: qcow2
flatten: true
regen_initramfs: true
emit_domain_xml: true
vm_name: migrated-test-vm
EOF

sudo h2kvmctl --config test-migration.yaml
```

```
virsh list --all           # VM should appear
virsh start migrated-test-vm # Start it
virsh console migrated-test-vm # Check console
```

POC Checklist

Validate these items to confirm migration readiness.

Test	How to Verify	Expected Result
☐ VMDK converts to qcow2	<code>qemu-img info out/*.qcow2</code>	Format: qcow2, size correct
☐ Linux VM boots on KVM	<code>virsh start vm-name</code>	Reaches login prompt
☐ Windows VM boots (no BSOD)	<code>virsh start win-vm</code>	Reaches desktop/login
☐ Network works post-migration	ping from migrated VM	Connectivity OK
☐ fstab is correct	<code>cat /etc/fstab</code> inside VM	UUID-based entries
☐ initramfs has virtio drivers	<code>lsinitrd</code> inside VM	virtio_blk, virtio_net present
☐ vSphere discovery works	TUI vSphere tab or <code>govc ls</code>	VMs listed with details
☐ Batch queue works	Select 3+ VMs, Ctrl+R	Sequential processing, results
☐ KubeVirt deploy works	<code>kubectl get vm</code>	VM created, PVC bound
☐ YAML automation works	<code>h2kvmctl --config test.yaml</code>	Reproducible results

POC VM Selection Guide

Start With

- Non-production Linux VM (CentOS/RHEL)
- Small disk (<20GB) for fast test
- Standard config (no GPU passthrough)
- Single disk, single NIC

Then Test

- Windows Server VM (VirtIO injection)
- Multi-disk VM (data disk preservation)
- UEFI boot VM (firmware detection)
- Production-like workload (DB, web server)

Common Issues & Fixes

Issue	Solution
govc: certificate error	Set <code>GOVC_INSECURE=1</code> or import vCenter CA cert
Permission denied on qcow2	Run with <code>sudo</code> — KVM needs root for disk access
VM boots to GRUB rescue	Enable <code>--regen-initramfs</code> in config
No network after boot	Check bridge config: <code>virsh net-list</code>
Windows BSOD	Add <code>--windows --virtio-win-iso /path/to/virtio-win.iso</code>